A VLM-Based System for Intracranial Hemorrhage Assessment on Brain CT

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Abstract

This project proposes a deep learning (DL) and vision-language model (VLM) pipeline for intracranial hemorrhage (ICH) assessment on non-contrast head CT. The system combines hemorrhage detection, subtype classification, segmentation, and mask-based volume estimation with evidence-grounded medical report generation.

Rather than relying on a VLM to interpret CT images and estimate hemorrhage characteristics independently, the proposed approach provides the VLM with structured findings derived from specialized image-analysis models. The research will investigate whether this combination improves the factual consistency and clinical completeness of generated reports.

1 Research Objectives

The project aims to develop a pipeline that:

1. Detects the presence of intracranial hemorrhage.
2. Classifies hemorrhage into five subtypes: epidural hemorrhage (EDH), subdural hemorrhage (SDH), subarachnoid hemorrhage (SAH), intraparenchymal hemorrhage (IPH), and intraventricular hemorrhage (IVH).
3. Segments hemorrhage regions and estimates their volumes.
4. Converts image-analysis results into structured findings.
5. Uses a VLM to generate a report grounded in the predicted findings, locations, and volumes.

The central hypothesis is that providing a VLM with structured, model-derived evidence will reduce unsupported statements compared with generating reports directly from CT images.

2 Proposed Pipeline

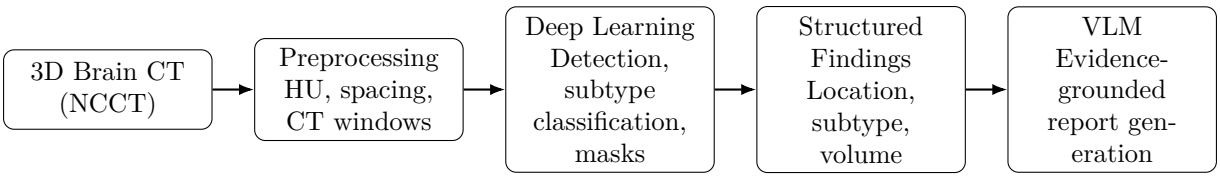


Figure 1: Overview of the proposed DL–VLM pipeline.

The system separates image analysis from language generation. Specialized DL models estimate the visual findings, while the VLM organizes these findings into a coherent report.

3 CT Data Representation and Preprocessing

3.1 3D CT to 2D Slices

A CT examination is a three-dimensional volume composed of ordered axial slices. For a 3D volume with dimensions $H \times W \times N$, each slice can be represented as a 2D image of size $H \times W$.

The approximate number of slices depends on scan coverage and reconstruction thickness:

- Thick slices of approximately 3–5 mm may contain around 20–50 slices per examination.
- Thin slices of approximately 0.5–1.25 mm may contain around 150–300 or more slices per examination.

These are illustrative ranges rather than fixed dataset properties. The actual number of slices must be obtained from each examination.

3.2 Preprocessing

The preprocessing stage should preserve the anatomical information required for hemorrhage detection and segmentation.

1. **Intensity handling:** convert stored pixel values to Hounsfield Units (HU) when the original metadata supports this conversion. Do not assume that every released image retains raw HU values.
2. **Spatial handling:** preserve orientation and voxel spacing, and resample volumes when required by the model.
3. **CT windowing:** use appropriate intensity windows. Multiple windows may help expose different hemorrhage characteristics. A subdural-sensitive window should also be investigated because thin SDH may be missed using only a conventional brain window.
4. **Slice preparation:** maintain slice order and spatial correspondence between images, predictions, and masks.

Diffusion-based motion-artifact correction may be evaluated as an optional preprocessing experiment. It should not be assumed to improve hemorrhage detection without downstream validation.

4 Dataset Plan

The following datasets are potential resources for training and evaluation. Their exact usable subsets, label formats, licenses, and image representations must be checked before implementation.

Dataset	Available supervision	Potential role and considerations
M3D-Cap	3D medical images paired with textual descriptions or reports.	Potential resource for medical image–text learning. It covers multiple anatomical regions; the brain CT subset must be identified and verified.
M3D-Seg	3D medical images with segmentation annotations in supported subsets.	Potential resource for segmentation and multimodal learning. Verify the availability of brain CT cases, hemorrhage masks, and associated text.
RSNA ICH	More than 25,000 CT examinations and 874,035 2D images, with image-level hemorrhage labels and five subtype labels.	Useful for classification and detection experiments. The original RSNA labels should not be treated as pixel-level segmentation masks.
CQ500	491 head CT examinations with radiologist assessments.	Potential resource for examination-level classification and external evaluation. The original dataset does not provide general-purpose pixel-level hemorrhage masks.
PhysioNet CT-ICH	A small collection of CT examinations with hemorrhage subtype labels and radiologist-delineated regions in the released data.	Potential resource for segmentation and evaluation. The collection is limited in size and focused on traumatic hemorrhage; verify the specific public release and available files.
BHSD	192 reconstructed CT volumes with pixel-level masks for five hemorrhage subtypes, as described in the surveyed study.	A particularly relevant candidate for multi-subtype segmentation. Verify the release, license, split definitions, and compatibility with the intended pipeline.

Dataset strategy: RSNA can support classification experiments, while BHSD and verified segmentation subsets can support mask prediction. M3D-Cap and M3D-Seg may support multimodal or segmentation experiments when their actual data and annotations match the task. CQ500 and PhysioNet CT-ICH may be used for evaluation when the labels and inclusion criteria are compatible.

Data splitting must be performed at the patient or examination level, not randomly by slice. Otherwise, adjacent slices from the same patient may appear in both training and test sets, leading to data leakage and overestimated performance.

5 Deep Learning Component

5.1 Detection and Subtype Classification

The classification module predicts whether hemorrhage is present and estimates the probability of each subtype:

$$\mathbf{p} = [p_{\mathrm{EDH}}, p_{\mathrm{SDH}}, p_{\mathrm{SAH}}, p_{\mathrm{IPH}}, p_{\mathrm{IVH}}].$$

Because multiple hemorrhage types may coexist in one examination, the classification task should support multi-label predictions where the dataset permits it.

Candidate architectures include CNNs and vision transformers. A 3D model can exploit inter-slice context, while a 2D model with slice-level aggregation may be easier to train when only 2D labels are available.

The model architecture should be selected through controlled experiments rather than assuming that one architecture is universally superior.

5.2 Segmentation

A segmentation model predicts a voxel-level mask:

$$M(x, y, z) \in \{0, 1\},$$

where $M(x, y, z) = 1$ indicates a voxel classified as hemorrhage. If subtype-specific masks are available, the system can additionally estimate the location and volume of each subtype.

A 3D segmentation architecture such as nnU-Net is a candidate baseline. Its suitability must be evaluated against the available training data, annotation quality, computational budget, and five-subtype requirements.

5.3 Hemorrhage Volume Estimation

Volume should be calculated from the predicted mask and physical voxel spacing rather than generated directly by the VLM.

For a binary mask, the estimated volume is

$$V_{\mathrm{mL}} = \frac{\sum_{x,y,z} \mathbb{1}[M(x, y, z) = 1]\, s_x s_y s_z}{1000},$$

where s_x, s_y, and s_z are the voxel spacings in millimeters.

The calculation is valid when the mask and spacing refer to the same spatial coordinate system. Subtype-specific volumes require subtype-specific masks or another validated method of separating the corresponding regions.

6 Structured Findings and VLM Report Generation

The DL outputs are converted into a structured representation before being passed to the VLM.

The structured input contains four evidence groups:

1. **Hemorrhage presence:** positive, negative, or uncertain.
2. **Subtype:** predicted subtype labels and their confidence scores.
3. **Location:** anatomical region, side, and spatial extent derived from validated predictions and masks.
4. **Volume:** mask-derived volume in milliliters, with uncertainty or a quality flag when appropriate.

A candidate generator is Qwen3-VL with parameter-efficient fine-tuning such as LoRA, subject to model availability and compatibility with the chosen data.

The VLM receives the structured findings and generates a report that describes the supported observations. It should not invent a subtype, anatomical location, or volume that is absent from the input

evidence. Uncertain predictions should remain explicitly uncertain rather than being converted into definitive statements.

The VLM's role is therefore to organize and communicate findings, not to replace the dedicated detection, classification, or segmentation models.

7 Related Work: Seven-Paper Review

The surveyed studies can be grouped into three categories: (1) direct MLLM assessment of CT images, (2) segmentation and artifact correction, and (3) classifier-guided VLM report generation.

7.1 Direct MLLM Assessment of CT Images

Paper 1: Wang et al., Brain Hemorrhages (2025). *Zero-shot multi-modal large language models v.s. supervised deep learning: A comparative analysis on CT-based intracranial hemorrhage subtyping.*

Method: The study compared six zero-shot MLLMs with four supervised DL models on 192 reconstructed CT volumes from the BHSD dataset. The evaluation used 561 randomly selected 2D slices. The MLLMs were asked to identify hemorrhage presence, subtype, location, and other findings.

Key results: Gemini 2.0 Flash achieved a macro-F1 of approximately 0.31 for subtype classification, while Swin-v2-B achieved approximately 0.86. Swin-v2-B also achieved 0.9091 accuracy for hemorrhage presence in the reported experiment.

Limitations: The MLLMs were evaluated zero-shot, while the DL models were trained for the task. The test split was performed by slice rather than patient, creating a potential data-leakage risk. The study therefore supports specialized DL as a strong baseline but does not establish that every fine-tuned VLM will perform poorly.

Implication: Use specialized DL for the core image-analysis tasks and evaluate the VLM separately for report generation.

Paper 2: Koyun et al., Diagnostics (2025). *Detection of Intracranial Hemorrhage from Computed Tomography Images: Diagnostic Role and Efficacy of ChatGPT-4o.*

Method: A retrospective study of 240 examinations, including 120 ICH cases and 120 controls. ChatGPT-4o received five radiologist-selected consecutive slices per examination and answered structured diagnostic questions.

Key results: The first evaluation reported 68.3% accuracy, 79.2% sensitivity, and 57.5% specificity for hemorrhage detection. Correct subtype identification was reported for 34% of hemorrhagic regions, while fully correct localization was only 7.3%. Agreement between two evaluation sessions was moderate ($\kappa = 0.469$).

Limitations: Only five selected slices were supplied per examination. The selection favored visible hemorrhage regions, and conversion to JPEG may have altered image information. The cohort came from one hospital and excluded some difficult cases.

Implication: Evaluate subtype, localization, and repeatability independently. Do not treat a correct positive/negative answer as evidence that the full report is correct.

Paper 3: Wu et al., Cureus (2026). *Where Are We Now? Benchmarking Large Language Models (LLMs) in Computed Tomography (CT)-Based Detection of Intracranial Hemorrhage.*

Method: Three web-based MLLMs were evaluated on 400 RSNA slices from 400 patients: 200 normal images and 200 hemorrhagic images, with 40 images for each subtype. The task used a zero-shot prompt.

Key results: Reported accuracy was 61.0% for Grok-2, 62.3% for Gemini 1.5 Flash, and 59.3% for ChatGPT-4o. All three models showed subtype-specific weaknesses, and SAH was difficult to identify.

Limitations: The experiment used single PNG slices rather than full CT examinations, with no access to adjacent-slice context or the original DICOM intensity range. Only cases with a single subtype were included. The reported abstract and table contain a small discrepancy; the values above follow the table.

Implication: Evaluate models on complete examinations where possible and report per-subtype performance rather than accuracy alone.

Paper 4: Shankar et al., Cureus (2026). *Diagnostic Role of ChatGPT in the Detection of Intracranial Hemorrhage on Non-contrast Computed Tomography: A Retrospective Case-Control Study.*

Method: The study evaluated GPT-5.3 on 276 examinations, comprising 138 hemorrhage cases and 138 controls. The model received four to six preselected slices per examination and a structured prompt.

Key results: The reported sensitivity was 89.9%, specificity 84.8%, accuracy 87.3%, and Cohen's $\kappa = 0.75$. Performance varied by subtype; SDH sensitivity was 61.5%, and EDH sensitivity was 66.7%.

Limitations: The study used a balanced, single-center case-control cohort. Radiologists preselected the slices, potentially making the task easier than automated full-volume assessment. Four subtypes were reported, without IVH results. The study's nonsignificant McNemar test does not establish equivalence to radiologists.

Implication: Structured prompts may improve MLLM performance under selected conditions, but full-volume automation and subtype-specific evaluation remain necessary.

7.2 Classifier-Guided Report Generation

Paper 5: Yang et al., MICCAI 2026. *Entity-Aware Head CT Report Generation via Classification-Derived Label Priors.*

Method: The HEADLINE challenge system extracted normalized clinical entities from reports, trained a classifier to predict label priors, and supplied the predicted labels to a VLM for report generation. The study used 14,132 reports for the Findings component and 16,468 for the Assessment component, with 100 selected labels.

Key results: On 500 internal cases, the ensemble system improved the reported S_{auto} score from 0.1914 without labels to 0.2122 with predicted labels. Using ground-truth labels achieved 0.2293, indicating a gap between predicted and ideal structured evidence. The system ranked fifth in the reported validation round.

Limitations: Incorrect predicted labels can propagate into incorrect report statements. The label representation does not fully encode temporal changes, and text similarity metrics may not reliably reflect clinical correctness. The system does not provide hemorrhage segmentation or mask-based volume estimation.

Implication: This is the closest architectural baseline to the proposed classifier-to-VLM approach. The proposed work will investigate whether segmentation-derived location and volume add useful evidence beyond classification labels.

7.3 Segmentation and Artifact Correction

Paper 6: Ironside et al., Stroke (2019). *Fully Automated Segmentation Algorithm for Hematoma Volumetric Analysis in Spontaneous Intracerebral Hemorrhage.*

Method: A CNN-based system was trained on 260 examinations and tested on 40 examinations from a cohort of 300 patients with spontaneous supratentorial intraparenchymal hemorrhage.

Key results: The reported Dice score was 0.894 ± 0.264 against manual segmentation and 0.905 ± 0.254 against semi-automated segmentation. The estimated volume showed strong agreement with manual measurement ($R^2 = 0.981$).

Limitations: The study focused on spontaneous intraparenchymal hemorrhage, not all five hemorrhage subtypes. The test set was small and external validation was identified as necessary. These results cannot be assumed to generalize to EDH, SDH, SAH, or IVH.

Implication: Use mask-based volume estimation, but validate each subtype separately and report volume errors in addition to Dice.

Paper 7: Chen et al., AMAI (2024). *Head CT Scan Motion Artifact Correction via Diffusion-Based Generative Models.*

Method: The proposed HM-EDM method used conditional diffusion to correct motion artifacts in reconstructed head CT images. Training pairs were generated by simulating motion corruption, and evaluation included simulated data, a phantom, and real mobile CT images.

Key results: The reported summary indicates improved image quality compared with CNN and standard DDPM baselines, with radiologist assessments suggesting improved diagnostic value. Exact quantitative metrics should be verified from the full paper before citation.

Limitations: The method addresses motion artifacts rather than general CT denoising. Simulated artifacts may differ from real-world artifacts, and image restoration does not guarantee improved hemorrhage detection. Generative restoration may also remove small lesions or introduce misleading details.

Implication: Treat diffusion correction as an optional ablation, not a required pipeline component. Measure downstream detection and segmentation performance, especially for small hemorrhages.

8 Research Gap

The reviewed studies suggest several limitations in existing approaches:

1. **Direct MLLM interpretation remains unreliable.** The surveyed studies report variable detection performance, weaker subtype classification, localization errors, and inconsistent outputs across evaluation settings.
2. **Classification labels alone provide limited evidence.** Classifier-guided report generation improves text-generation metrics, but incorrect labels can propagate into incorrect statements.
3. **Segmentation and report generation are often evaluated separately.** Strong mask-based volume estimation does not automatically produce a clinically coherent report, while report-generation metrics do not establish correct lesion localization or volume.
4. **Artifact correction is not equivalent to clinical improvement.** Better image similarity scores do not prove better hemorrhage detection or preservation of small lesions.

Proposed direction: Integrate detection, subtype classification, segmentation, and mask-derived volume estimation into a structured evidence layer before VLM report generation. Evaluate whether this additional evidence improves report factuality and consistency compared with a VLM that receives CT images or classification labels alone.

This is a research hypothesis, not an established novelty claim. Its contribution must be demonstrated through controlled experiments against relevant baselines.

9 Experimental Design

9.1 Evaluation Metrics

Component	Metrics
Hemorrhage detection	Sensitivity, specificity, AUROC, AUPRC, and patient-level classification performance.
Subtype classification	Per-subtype sensitivity and specificity, macro-F1, and confusion matrix.
Segmentation	Dice score, HD95, and lesion-level sensitivity, including separate results for small lesions.
Volume estimation	Mean absolute error (MAE) and root mean squared error (RMSE) in mL, plus agreement with reference volume.
Report generation	Clinical entity accuracy, factual consistency, unsupported-claim rate, and text similarity metrics.
Reliability	Repeatability across runs and the rate of uncertain or contradictory statements.

Text similarity scores should not be used as the sole evidence of clinical correctness. A report can resemble a reference report while still containing an incorrect subtype, location, or volume.

9.2 Ablation Studies

The following experiments will test the contribution of each component:

1. **VLM baseline:** generate a report directly from the available CT input.
2. **Classification-guided VLM:** provide predicted hemorrhage presence and subtype labels.
3. **Full structured-evidence VLM:** additionally provide segmentation-derived location and volume.
4. **Oracle evidence:** provide reference labels and reference findings where available, to estimate the upper bound of report generation with correct evidence.
5. **Preprocessing ablation:** compare the baseline preprocessing pipeline with and without optional diffusion correction, if implemented.

The oracle experiment is an analytical upper-bound comparison, not a deployable system configuration.

All experiments should use patient-level splits. External evaluation should be performed when compatible datasets and labels are available. Results should be reported separately for each subtype, with special attention to small lesions, SDH, EDH, and SAH.

10 Expected Contributions

The expected contributions are:

- A modular DL–VLM pipeline for five-subtype ICH assessment.
- Mask-based volume estimation linked to physical voxel spacing.
- A structured evidence representation connecting image predictions to report generation.

- A controlled comparison of direct VLM generation, classification-guided generation, and generation using segmentation-derived evidence.
- An evaluation framework covering detection, subtype classification, segmentation, volume accuracy, and report factuality.

The main contribution is not simply combining DL and a VLM, but testing whether explicit spatial and quantitative evidence improves the reliability of generated reports.

11 Implementation Plan

1. **Dataset audit:** verify available files, label definitions, licenses, HU representation, voxel spacing, and patient identifiers.
2. **Baseline models:** establish detection and subtype-classification performance using compatible labels.
3. **Segmentation:** train and evaluate mask prediction on datasets with verified pixel- or voxel-level annotations.
4. **Volume validation:** compare mask-derived volume against available reference measurements.
5. **Structured evidence:** convert model outputs into a consistent schema containing presence, subtype, location, and volume.
6. **VLM integration:** fine-tune or adapt the selected VLM to generate reports from structured findings.
7. **Ablation and validation:** compare the proposed system with direct VLM generation and classification-guided generation, including external evaluation when possible.

12 Conclusion

This proposal separates CT image analysis from medical report generation. Specialized DL models estimate hemorrhage presence, subtype, location, segmentation, and volume; the VLM then converts these structured findings into a readable report.

The literature motivates this division of responsibilities, while also showing that model performance depends on dataset composition, slice selection, subtype, and evaluation design. The key experiment is therefore to determine whether segmentation-derived spatial and quantitative evidence improves report reliability beyond classification-only guidance.